Homework Cheatsheet

HW1 Basic Knowledge

1. Overfitting Judgment

- "Overfitting occurs only when training error is zero." **Wrong**.
- Overfitting: low training error, poor test performance.

2. KNN K Value and Variance

- Larger K: **variance decreases** (smoother prediction).

3. Bayes Classifier Expected Error

Expected error rate:

$$E[1 - \max_j q_j(X)]$$

$q_j(x) = P(Y = j|X = x)$.

4. Residual Sum in SLR/MLR

- SLR & MLR (with intercept): $\sum e_i = 0$.

5. R-squared vs Adjusted R-squared

- Same number of parameters: R^2 or Adjusted R^2 equivalent for comparison.

6. Linear vs KNN Regression

- Linear: simple, interpretable, high bias.
- KNN: flexible, non-parametric, high variance, computationally expensive.

HW2 Basic Knowledge

1. Logistic Log-Likelihood Concavity

- Log-likelihood is concave (Hessian negative semi-definite).

2. Logistic MLE Algorithms

- IRLS (Iteratively Reweighted Least Squares).
- Advantage: closed-form per iteration, fast convergence.
- Disadvantage: high computational cost (matrix inversion).

3. Logistic Continuous Coefficient

- β_i: unit increase in X_i changes log-odds by β_i.

4. Logistic ROC Pseudocodes

- Fit model = LogisticRegression().fit(X$_train, y_train)y_prob = model.predict_proba(X_train)[:, 1]$
- fpr, tpr, $_= roc_curve(y_train, y_prob) Plot fpr vs tpr.$

5. LDA vs Logistic

- LDA: generative, models $P(X|Y)$, assumes Gaussian.
- Logistic: discriminative, directly models $P(Y|X)$.

6. LDA Discriminant Function

$$\delta_k(x) = x^\top \Sigma^{-1} \mu_k - \frac{1}{2} \mu_k^\top \Sigma^{-1} \mu_k + \log \pi_k$$

7. Pooled Covariance Unbiased

- $\hat{\Sigma} = \frac{1}{n-K} \sum (X_i - \hat{\mu}_k)(X_i - \hat{\mu}_k)^\top$: unbiased for Σ.

8. QDA Discriminant Function

$$\delta_k(x) = -\frac{1}{2}(x - \mu_k)^\top \Sigma_k^{-1} (x - \mu_k) - \frac{1}{2} \log |\Sigma_k| + \log \pi_k$$

9. Multiclass Confusion/ROC

- Confusion: $K \times K$ matrix (rows true, cols predicted).
- ROC: One-vs-Rest (OvR) or One-vs-One (OvO) AUC.

10. CV Folds K Effect

- Larger K: bias ↓, variance ↑ (closer to full data).
- Smaller K: bias ↑, variance ↓.

HW3 Basic Knowledge

1. Cubic Spline as Constrained LR

- Use truncated power basis/B-spline + continuity constraints on value, 1st/2nd derivatives at knots.

2. Piecewise vs Local Polynomial

- Piecewise: fixed global regions.
- Local: weighted LS around each target point.

3. Bandwidth h in Local Regression

- Larger h: bias ↑, variance ↓.

4. Regression Tree Similarity

- Most similar to piecewise constant regression.

5. Finer Tree Bias/Variance

- Finer tree: bias ↓, variance ↑.

6. Lowest Bias Method

- Natural splines (most flexible).

7. Random Forest Variable Importance

- VIM: average total decrease in RSS/Gini across all trees.

HW4 Basic Knowledge

1. Maximal Margin Classifier Equivalence

Equivalent form:

$$\min \frac{1}{2} \|\beta\|^2 \quad \text{s.t.} \quad y_i(x_i^\top \beta + \beta_0) \geq 1$$

2. Linear SVC Representation

- $f(x) = \beta_0 + \sum a_i \langle x_i, x \rangle$.
- Only support vectors have $a_i > 0$.

3. RBF γ Effect

- Larger γ: bias ↓, variance ↑ (more local).
- Smaller γ: bias ↑, variance ↓.

4. PCA Missing Data & Preprocessing

- Missing: impute or delete (no direct handling).
- Best: center + standardize if scales differ.

5. K-means++

- Initialization: select centers far apart probabilistically.
- Advantage: better convergence, avoids poor local minima.